

Riya Bhart

riyabhart02@gmail.com | +92 3213990311 | LinkedIn: [riya-bhart-339036287](https://www.linkedin.com/in/riya-bhart-339036287) | GitHub: <https://github.com/RiyaBhart/>

EDUCATION

FAST NUCES Karachi

B.S. in Artificial Intelligence

- o **Concentrations:** Artificial Intelligence & Machine Learning Systems and Engineering
- o **CGPA:** 3.26/4.00
- o **Related Coursework:** Machine Learning, Artificial Intelligence, Database, Knowledge Representation & Reasoning, Data Structures & Algorithms, Design Analysis & Algorithm, Object-Oriented Programming, Statistics & Applications, Linear Algebra, Calculus, Operating System, Computer Organization & Assembly Language

SKILLS

Programming: Python, SQL, C++, C, MASM

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, FastAPI, React, Ultralytics (YOLOE), Captum, spaCy, NetworkX, SPARQL/OWL (Knowledge Graphs), Pandas, NumPy, Matplotlib, pyGame

AI/ML Techniques: Prompt Engineering, Embeddings & Semantic Search, Object Detection, Visual Transformers (ViT), Self-Supervised Learning, Explainable AI (XAI), Ontology-based Reasoning, Rule-based Inference (SWRL)

Tools: Docker, Git, PostgreSQL, Jupyter Notebook, Streamlit, AI2-THOR, EasyOCR, pytsx3, Excel

PROJECTS

[GitHub](#)

MediFlow — AI-Powered Clinic Operations Platform - RAG • Multi-Agent • MLOps • AIOps • DevOps

- Engineered a full MLOps pipeline using **MLflow** for experiment tracking, model registry, and champion/challenger promotion; automated prediction logging, drift detection, and retraining via **Celery Beat** workers, closing a continuous loop from live inference to model refresh without manual intervention.
- Containerized the full backend stack with **Docker Compose** across isolated services (API, Redis, Celery worker, Celery Beat, ML service) and built a **GitHub Actions** CI/CD pipeline that auto-deploys to **AWS EC2** on push to main, runs Alembic migrations, refreshes RAG ingestion, and rebuilds the stack reproducibly.
- Implemented an **AIOps** monitor agent that reasons over live operational signals — API health, worker availability, model drift, booking volume, and STT/TTS latency — classifying incident severity, triggering alerts, and initiating retraining flows to keep the production stack self-healing.
Tech: Python, FastAPI, Docker, Docker Compose, AWS EC2, GitHub Actions, Celery, MLflow, PostgreSQL, Redis, Alembic, Caddy, Vercel

Vision-to-Voice — Predictive Navigation Assistant for Visually Impaired Students - DINOv2 • JEPA • Scene MLP • XAI

- Built an end-to-end visual navigation pipeline using a frozen DINOv2 ViT-S/14 backbone and a JEPA-lite world model (held-out MSE: 0.107) to predict future scene states and bias frontier exploration via surprise scores, achieving 21% environment coverage vs 9.5% with random selection
- Trained a Scene Context MLP on DINOv2 embeddings for socially-aware navigation (**89.89% accuracy**), eliminating inappropriate navigation actions entirely in ablation (1.95 → 0.00 per 10 steps); integrated conditional YOLOE open-vocabulary object detection triggered selectively on high-surprise frames (20.5% trigger rate).
- Implemented transformer interpretability via AttnLRP and SmoothGrad (Captum) with KMeans clustering over 100+ attribution maps to audit for shortcut patterns; built a real-time topological cognitive map with EasyOCR semantic landmark labeling, producing 82 nodes over 200 steps.
Tech: Python, PyTorch, DINOv2, YOLOE, EasyOCR, Captum, NetworkX, AI2-THOR, pytsx3

AIGES — Agentic AI Risk & Lending Engine - Reinforcement Learning (PPO) • LightGBM • Basel III Optimization

- Developed an autonomous credit decisioning system combining LightGBM-based PD models with a PPO reinforcement learning agent to optimize loan approvals.
- Trained dual Retail & SME risk models and a policy agent (200K timesteps) to maximize portfolio profit while minimizing default risk under Basel III capital constraints.
- Achieved **10× higher profit** and **~80% lower default rate** compared to rule-based and static PD-threshold strategies.

- Integrated regulatory capital calculations (RWA, capital requirements) and LLM-based explainability for transparent, regulator-aligned decisions.

Tech: Python, LightGBM, Stable-Baselines3 (PPO), Pandas, NumPy, Basel III Framework, LLM Explainability

LAYR — Machine Learning based Fashion E-commerce System - Recommendation • NLP • Time Series

Aug 2025 – Dec 2025

- Designed and deployed a comprehensive ML pipeline spanning recommendation systems, review analytics, demand prediction, and customer segmentation.
- Implemented a hybrid recommendation engine combining collaborative filtering, content-based filtering, and popularity-based ranking.
- Leveraged BERT for sentiment analysis and topic modeling on customer reviews.
- Applied ARIMA and Prophet models for category-level demand forecasting and inventory risk assessment.
- Conducted RFM-based customer segmentation and funnel analysis to enable data-driven personalization strategies.

Tech: Python, PyTorch, scikit-learn, Sentence-BERT, Prophet, PostgreSQL, FastAPI, MLflow.

Prodigy: AI Productivity Companion — LLMs • Full-Stack Web App

Feb 2024 – June 2024

- Built an AI-powered productivity platform integrating task management, Pomodoro workflows, smart notes, and analytics in a unified system.
- Integrated GPT-4 APIs for session-based note summarization and AI-driven productivity insights.
- Designed real-time reminders, calendar synchronization, and focus tracking with Firebase backend infrastructure.

Tech: React, TypeScript, Tailwind CSS, Firebase, Firestore, OpenAI GPT-4 APIs.

Companion: Ontology Driven Mental Health Chatbot — KG • Ontology • SWRL Reasoning • XAI

Oct 2025 – Nov 2025

- Engineered an ontology-driven chatbot leveraging OWL, SWRL, and SPARQL to infer emotional states and symptom patterns through rule-based reasoning.
- Designed causal “why” explanations and safety alert mechanisms for detecting high-risk indicators.
- Integrated NLP-based entity extraction with a React-powered conversational interface for interactive user engagement.

Tech: Python, FastAPI, React, OWL, RDF, SPARQL, spaCy.

EXPERIENCE

Team Leader Marketing & PR

AIESEC In Karachi

Spring 2026

- Led the Marketing & Public Relations function for AIESEC in Karachi, ensuring consistent brand visibility across all platforms.
- Managed and coordinated social media content, improving engagement and outreach across key channels.
- Executed branding, design, and communication plans, ensuring they aligned with AIESEC’s global standards.